The sequence $\sqrt{k}/k!$ is a multiplier sequence

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Abstract

$\{\sqrt{k}/k!\}_{k\geq 0}$ is a multiplier sequence; its exponential generating function has simple negative nonzero zeros with explicit quadratic localization. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

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1 Statement and argument

The answer to the question of Csordas and Forgács [1] is affirmative. We prove the stronger assertion that the associated entire function has only simple negative zeros, apart from its simple zero at the origin, and locate every zero between two explicit bounds.

Theorem 1.1. *The sequence $\gamma_0 = 0$, $\gamma_k = \sqrt{k}/k!$ for $k \geq 1$, is a classical multiplier sequence. More precisely, write*

$$\Phi(w) = \sum_{k=1}^{\infty} \frac{\sqrt{k}}{(k!)^2} w^k.$$

Its zeros are 0 and $-\lambda_j$, $j \geq 1$, all simple, where

$$\frac{j^2\pi^2}{4} < \lambda_j < \frac{(j+1)^2\pi^2}{4}.$$

In particular Φ belongs to the Laguerre–Pólya class of type I, with nonnegative Taylor coefficients.

1.1 A cosine subtraction lemma

Lemma 1.2. *Let μ be a nonatomic probability measure on $(0,1)$. The entire function*

$$F(z) = \cos z - \int_{(0,1)} \cos(tz)\, d\mu(t)$$

has only real zeros. Its zero at 0 has multiplicity two; in each interval $(j\pi, (j+1)\pi)$, $j \geq 1$, it has exactly one simple zero. The remaining zeros are the negatives of these zeros.

Proof. Local uniform domination proves that the integral is entire. The function F is even and has real Taylor coefficients. Since $0 < \int t^2\, d\mu(t) < 1$,

$$F(z) = -\frac{1 - \int t^2\, d\mu(t)}{2} z^2 + O(z^4),$$

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

so the zero at 0 is exactly double.

For a nonzero integer j and $z = j\pi + iy$,

$$|\cos z| = \cosh y, \qquad \left|\int \cos(tz)\, d\mu(t)\right| \le \int \cosh(ty)\, d\mu(t).$$

When $y \neq 0$ the second quantity is strictly less than $\cosh y$, because $0 < t < 1$. When $y = 0$ use instead

$$\left|\int \cos(j\pi t)\, d\mu(t)\right| \le \int |\cos(j\pi t)|\, d\mu(t) < 1.$$

Here the equality set of $|\cos(j\pi t)| = 1$ in $(0,1)$ is finite and has zero μ-measure.

On a horizontal line $z = x \pm iR$, uniformly in real x,

$$|\cos z| \ge \sinh R, \qquad \left|\int \cos(tz)\, d\mu(t)\right| \le \int \cosh(tR)\, d\mu(t).$$

Moreover

$$\frac{\int \cosh(tR)\, d\mu(t)}{\sinh R} \longrightarrow 0 \quad (R \longrightarrow \infty).$$

Indeed, the integrand ratio tends to zero for every $t < 1$ and is bounded by $\coth 1$ for $R \ge 1$. Dominated convergence applies.

For every sufficiently large R, Rouché's theorem on the rectangle with vertical sides $j\pi$ and $(j+1)\pi$, $j \ge 1$, and horizontal sides at imaginary parts $\pm R$ therefore gives exactly one zero of F, counted with multiplicity: this is the zero count for $\cos z$. Conjugation invariance forces this zero to be real, and it is simple. Letting R increase excludes any other zero in the strip. On the central rectangle with vertical sides $-\pi$ and π, the same argument gives two zeros counted with multiplicity, already accounted for by the double zero at 0. Evenness handles the negative strips. The strict vertical-side estimates exclude zeros on their boundaries. These regions cover the plane. □

1.2 A moment representation for a gamma ratio

Lemma 1.3. *For $x > 0$ define*

$$W(x) = \sqrt{x}\,\frac{\Gamma(x+\frac{1}{2})}{\Gamma(x+1)}.$$

There is a nonatomic probability measure μ on $(0,1)$ such that

$$W(x) = 1 - \int_{(0,1)} t^{2x}\, d\mu(t), \qquad x > 0.$$

Proof. For $s > 0$ put

$$\begin{aligned} K(s) &= \frac{1}{\pi}\int_0^s \frac{dt}{\sqrt{s-t}\sqrt{e^t-1}} \\ &= \frac{1}{\pi}\int_0^1 \frac{1}{\sqrt{1-u}}\sqrt{\frac{s}{e^{su}-1}}\, du. \end{aligned}$$

Since $e^{su} - 1 \ge su$, the integrand is bounded by the integrable function $\pi^{-1}[u(1-u)]^{-1/2}$. For fixed $0 < u < 1$, $s/(e^{su}-1)$ is strictly decreasing in s: the sign of its derivative is the sign of $e^v(1-v) - 1 < 0$, where $v = su > 0$. Thus K is continuous and strictly decreasing. Dominated convergence gives

$$K(0+) = \frac{1}{\pi}\int_0^1 [u(1-u)]^{-1/2}\, du = 1, \qquad K(\infty) = 0.$$

It follows that K is the survival function of a nonatomic probability measure ν on $(0,\infty)$, so that $\nu((s,\infty)) = K(s)$.

Tonelli's theorem, the gamma integral, and the beta integral show that

$$\begin{aligned}\int_0^\infty e^{-xs}K(s)\,ds &= \frac{1}{\pi}\left(\int_0^\infty e^{-xr}r^{-1/2}\,dr\right)\left(\int_0^\infty e^{-xt}(e^t-1)^{-1/2}\,dt\right)\\ &= \frac{1}{\pi}\frac{\sqrt{\pi}}{\sqrt{x}}B\left(x+\tfrac12,\tfrac12\right)\\ &= \frac{1}{\sqrt{x}}\frac{\Gamma(x+\frac12)}{\Gamma(x+1)}.\end{aligned}$$

For the second integral the substitution $q=e^{-t}$ gives $\int_0^1 q^{x-1/2}(1-q)^{-1/2}\,dq$. The survival-function identity, also directly obtained by Tonelli, is

$$1-\int_0^\infty e^{-xs}\,d\nu(s) = x\int_0^\infty e^{-xs}K(s)\,ds = W(x).$$

Push ν forward by $s\mapsto e^{-s/2}$. The resulting measure μ is nonatomic, has total mass one, is supported in $(0,1)$, and has the required moment identity. □

1.3 Identification and factorization

Let μ be the measure just constructed. Termwise integration of the locally uniformly convergent cosine series and the gamma duplication formula give

$$\begin{aligned}\frac{1}{\sqrt{\pi}}\left(\cos z-\int\cos(tz)\,d\mu(t)\right) &= \frac{1}{\sqrt{\pi}}\sum_{k=1}^\infty\frac{(-1)^kW(k)}{(2k)!}z^{2k}\\ &= \sum_{k=1}^\infty\frac{\sqrt{k}}{(k!)^2}\left(-\frac{z^2}{4}\right)^k = \Phi\left(-\frac{z^2}{4}\right).\end{aligned}$$

The coefficient identity here is

$$\frac{\Gamma(k+\frac12)}{\sqrt{\pi}\,\Gamma(k+1)} = \frac{(2k)!}{4^k(k!)^2}.$$

The cosine subtraction lemma now implies that all zeros of Φ are nonpositive real numbers. In fact every complex number w has a square root z with $w=-z^2/4$, and if $\Phi(w)=0$ then the lemma forces this z to be real. Multiplicities and the interval bounds in the theorem follow from the same substitution.

For completeness, Φ has order at most $1/2$. For $r\ge0$,

$$M_\Phi(r)\le\sum_{k\ge1}\frac{kr^k}{(k!)^2} = \sum_{k\ge1}\frac{r^k}{(k-1)!k!}\le\sqrt{r}\,e^{2\sqrt{r}}.$$

The last inequality follows by comparison with the product of two exponential series. Hadamard factorization for an entire function of order less than one therefore gives

$$\Phi(w) = w\prod_{j=1}^\infty\left(1+\frac{w}{\lambda_j}\right).$$

The factor in front is exactly w because $\Phi'(0)=1$. The product converges locally uniformly, also immediately from $\lambda_j>j^2\pi^2/4$. Its finite products have only nonpositive zeros and nonnegative coefficients. Hence Φ is in the required Laguerre–Pólya class. The Pólya–Schur characterization, in the normalization $\Phi(w)=\sum\gamma_kw^k/k!$, proves that $\{\sqrt{k}/k!\}_{k\ge0}$ is a multiplier sequence.

References

[1] G. Csordas and T. Forgacs, *Multiplier sequences, classes of generalized Bessel functions and open problems*, J. Math. Anal. Appl. **433** (2016), 1369–1389. https://arxiv.org/abs/1505.00355v2. DOI: https://doi.org/10.1016/j.jmaa.2015.08.047.